

Response to Reviewer 1

“The Seifert–van Kampen Theorem via Computational Paths”

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We thank the reviewer for the careful and constructive report. We retained the paper’s structure and addressed each identified location directly. The Lean development now includes proofs of both glue-naturality forms, the obstruction to an amalgamation-only SVK target, the full-target theorem for the global rewrite quotient, the low-degree π_0 tail of the fibration sequence, and a general sphere loop-quotient theorem.

The revision also makes the quotient being classified explicit in every nontrivial application. The relation generated by all constructors of the global `Path.Step` system makes each `PathRwQuot` loop fiber contractible.

To obtain a non-collapsing fundamental-group theorem without relabeling a normal-form quotient, we added `PresentedFundamentalGroup.lean`. A presented path space has proof-relevant generating edges; raw paths are freely generated by reflexivity, reversal, and composition; and homotopy is generated only afterward from named relators and the groupoid laws. Its fundamental group is the automorphism group of the basepoint in the resulting strict groupoid. The new theorem `presentedSeifertVanKampenGroupEquiv` constructs encode and decode, proves both round trips, and records preservation of multiplication, identity, and inversion:

$$\pi_1^{\text{pres}}(\mathcal{P}(G_1, G_2; H), *) \cong G_1 *_H G_2.$$

We instantiate it for the figure-eight and separately prove a nontrivial generator. The circle presentation is likewise proved equivalent to $\mathbb{Z}$. At the fundamental-group level, we additionally prove that $\mathbb{R} \rightarrow \text{AddCircle } 1$ is a covering map, classify Mathlib’s topological fundamental group of the circle by winding, and prove the group equivalence `presentedCircleTopologicalPiOneGroupEquiv`. A general geometric realization is also constructed from the nerve, and `Presented.Realization.hoNerveIso` recovers the original presented groupoid as its simplicial homotopy category. The final comparison with the topological fundamental groupoid of that realization, including the figure-eight, remains open and is stated as such.

Overall remarks

Reviewer comment. *The assumption bookkeeping is not applied consistently, and the reader must consult several separated tables and appendices to determine what is proved.*

Response. We consolidated the bookkeeping in three coordinated places: the main presented theorem prints its algebraic data directly, the separate global-rule schema prints all encode type-classes in its statement, Section 8 contains one manifest for those conditional parameters, and Appendix A gives, for every headline result, the exact Lean name, module, object classified, and

status (“presented fundamental group”, “global-rule quotient”, “conditional”, “completed presentation”, or “open”). The text explains that the amalgamation-only relation preserves word length and cites the formal impossibility theorems for that target.

Reviewer comment. *Claims about decidability and the “36 axioms” count are weaker than the headline presentation suggests.*

Response. Both claims were corrected. Arbitrary `RwEq` is no longer described as decidable without qualification: the paper now states that expression-level termination and confluence are proved, while path-level decidability requires an explicitly supplied `CompleteStepSystem`. The “36 axioms” headline and optional axiom-import inventory were removed. The source contains zero custom axiom declarations, as checked by `scripts/check_invariants.py` and `paper/svk/check_stats.py`.

Reviewer comment. *A key naturality lemma is asserted without proof.*

Response. We added a complete derivation in the paper, together with the two Lean theorems `Pushout.glue_natural_square_rweq` and `Pushout.glue_natural_rearranged_rweq`. Both are proved in the module `Path/CompPath/PathCompPath.lean`.

Detailed comments

Reviewer comment. *p. 2, Section 1, item 2: “ $\sim$ is decidable via normalization” conflicts with the metatheory table.*

Response. Item 2 states that the normalization function is a sound invariant, not an unconditional complete characterization. It identifies the proved abstract-expression fragment and points to `CompleteStepSystem.decideRwEq` for conditional decidability.

Reviewer comment. *p. 10, Section 2.4: confluence is marked “typeclass”, but it is not stated who discharges it.*

Response. The table now distinguishes two levels. The library discharges expression termination and confluence through `instHasTerminationProp` and `instHasConfluencePropExpr`. A caller must supply `HasJoinOfRw/HasConfluenceProp` at the trace-carrying `Path` level; no global instance is claimed.

Reviewer comment. *p. 16, Lemma 3.4: no proof is given; add a proof or Lean pointer.*

Response. The added proof sets $L = \text{inl}_*(f_*(p))$ and $R = \text{glue}(c_1) \cdot \text{inr}_*(g_*(p))$, uses the stored witness $L^{-1} \cdot R \sim \text{glue}(c_2)$, and then applies left congruence, associativity, inverse cancellation, and the unit law. The paper cites `glue_natural_square_rweq`.

Reviewer comment. *p. 16, Corollary 3.5: state explicitly that it follows by right-multiplying by $\text{glue}(c_2)^{-1}$.*

Response. The proof now says exactly this, including reassociation and inverse cancellation. The corresponding Lean theorem is `glue_natural_rearranged_rweq`.

Reviewer comment. *p. 21, Section 5.1: Theorem 5.1 gives no indication that it is conditional.*

Response. The revised Section 5 now begins with an unconditional theorem for presented computational path spaces. Its raw paths and generated homotopy are defined before `encode`, and Lean theorem `presentedSeifertVanKampenGroupEquiv` constructs both maps, proves both round trips, and packages the group homomorphism laws. Its hypotheses `GroupLaws` and `AmalgamationLaws` are explicit algebraic structures; the latter includes the source-group laws

and two `PreservesGroupOps` certificates. None of these are encode assumptions. For the circle, the separate theorem `presentedCircleTopologicalPiOneGroupEquiv` identifies the presented fundamental group with Mathlib’s topological `FundamentalGroup (AddCircle 1) 0`.

The separate theorem for the global-rule `PathRwQuot` is titled “Conditional global-rule SVK schema”, and all four required typeclasses are part of its statement. We also distinguish the full target, which includes free-group reduction, from the amalgamation-only target. Lean proves that round-trip and bijectivity interfaces for the latter are impossible: `hasPushoutSVKEncodeDecode_impossible` and `hasPushoutSVKDecodeAmalgBijection_impossible`. Separately, `seifertVanKampenFullEquiv_collapsed` proves the full-target equivalence for the global rewrite quotient and is explicitly labeled degenerate. The completed-expression theorem `scopedSeifertVanKampenEquiv` remains as a separate normal-form result rather than serving as the fundamental-group theorem.

Reviewer comment. *p. 27, Table 1: the sphere row says “SVK: Yes”, while Appendix A lists no dependency for S^2 .*

Response. We added the general Lean theorem `sphereN_piOne_equiv_unit`, specialized from `pathRwQuotLoopEquivUnit`. Table 1 and Appendix A agree: this global-rule loop-quotient result has no SVK encode/decode dependency and follows from the universal quotient collapse.

Reviewer comment. *p. 36, Definition 7.6: the long exact sequence is truncated at $\rightarrow 1$; should it continue through π_0 ?*

Response. Yes. The display continues

$$\pi_1(B) \rightarrow \pi_0(F) \rightarrow \pi_0(E) \rightarrow \pi_0(B).$$

The Lean development defines `boundaryPi0`, `fiberPi0ToTotalPi0`, and `totalPi0ToBasePi0`, and packages them with the pointed adjacent-composition laws in `LowDegreeFibrationSequence`; the constructor is `lowDegreeFibrationSequence`. The paper explains that a terminal 1 is shorthand only under path-connectedness assumptions.

Reviewer comment. *p. 5, related work: add Agda/HoTT-Agda SVK developments and Lean Mathlib’s fundamental groupoid material.*

Response. Section 1.2 discusses the machine-checked `VanKampen` development in `HoTT-Agda` and Mathlib’s classical fundamental groupoid/fundamental group modules, and explains how their objects differ from explicit equality-trace quotients.

Reviewer comment. *No repository link is provided.*

Response. The abstract and a dedicated Code Availability subsection now cite the immutable Lean-only Zenodo artifact <https://doi.org/10.5281/zenodo.21696875>. It contains the 21 paper entry modules, all transitive local Lean dependencies, pinned toolchain and Lake metadata, theorem manifest, license, and invariant checker. We retain <https://github.com/Arthur742Ramos/ComputationalPathsLean> as the ongoing development repository.

Reviewer comment. *p. 44, Section 8.6 Part C: optional axioms are excluded from the headline count and should be flagged next to each theorem.*

Response. The source contains no such optional axiom files, so Part C and the “36 axioms” count were removed rather than merely relocated. Each application is labeled locally as presented fundamental group, global-rule/proved, conditional, completed presentation, or open, and Appendix A repeats the same status.

Reviewer comment. *p. 36, path tactics: state what fraction of theorems rely on them.*

Response. We added a reproducible lexical count. None of 19,967 `theorem/lemma` declarations directly invokes a `path_*` tactic. Across all 53,451 declarations counted, including definitions, instances, and examples, 11 do so (approximately 0.021%); in the twenty-one paper modules, one of 1,018 declarations does so (0.098%). The paper therefore characterizes these tactics as convenience wrappers rather than a mature dominant automation layer.

Reviewer comment. *The seventh line of the table on p. 51 does not express Theorem 6.8.*

Response. The Klein-bottle row and the corresponding theorem now name the declaration `kleinBottleScopedPiOneEquivIntProd`, and identify its source as the scoped expression quotient in `ClassicalPresentationsScoped.lean`. Both places state that the quotient is classified by the $\mathbb{Z} \rtimes \mathbb{Z}$ normal form and do not identify it with the global-rule `PathRwQuot`.

Minor corrections

- The structure paragraph now includes Section 9 before Section 10.
- The decode subsection now begins “For pushouts, the decode map converts a word to a loop”, correcting the punctuation.
- The bullets in Example 6.12 begin on lines separate from the heading.
- “Part C” was removed with the optional-axiom inventory, eliminating the punctuation issue.
- Future Work now spells out the octonionic sequence $S^7 \rightarrow S^{15} \rightarrow S^8$.